Half-life and heavy tail in the disruption of three-body systems

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We measure the lifetime distribution of non-hierarchical three-body systems in 9,000 regularized planar integrations, with equal masses, with two light masses and one heavy mass, and with two heavy masses and one light mass, starting from rest and with virial ratio 0.5, up to ten thousand crossing times. The half-life, the median lifetime, ranges from about 3 to 30 crossing times and decreases with the mass disparity. The disruption, however, is not exponential: from 0.8% to 8% of the systems survive to one thousand crossing times, many orders of magnitude above the exponential prediction, and the survival tail approximately follows a power law. In the start from rest, the exponent lies between 0.63 and 0.94, close to the value 2/3 predicted for Lévy flights, and the long-lived systems are dominated by a single long excursion. In the virial start, the tail is heavier, with an exponent between 0.17 and 0.45, and, with a heavy pair, the long-lived systems are nearly stable hierarchical configurations, with hundreds of revolutions of the light body.

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I. INTRODUCTION

A non-hierarchical three-body system almost always ends with the ejection of one of its bodies [3]. A natural question is how long it lives before that happens. If the disruption were a memoryless process, with a constant probability of ejection per unit time, the fraction of surviving systems would fall exponentially, as in radioactive decay, and the half-life would summarize the entire statistics. Numerical studies have shown, however, that the early part of the lifetime distribution is approximately exponential but that the tail is algebraic, with survival falling as a power of time, and have attributed this tail to very long excursions of one of the bodies, the Lévy flights [1, 2].

This note measures the half-life and the tail of the distribution for three mass ratios and two types of initial condition, and classifies the long-lived systems according to the mechanism that keeps them alive.

II. METHOD

A. Simulations

We use the regularized integrator of the previous notes [4], with tolerance 10^{-12}, in planar simulations with $G = 1$ and without softening of the potential. The mass groups are (1, 1, 1), (1, 1, 10), and (10, 10, 1), with a start from rest and with virial ratio 0.5, under the same random initial conditions as note FR-TC-05 [5] and new seeds, in 1,500 runs per group and per set, for a total of 9,000. The integration limit was 10^4 crossing times t_{cr}. The lifetime T is the instant of ejection, defined as in note FR-TC-05; runs without ejection up to the limit are treated as censored. The relative energy error remained below 4×10^{-7} in all runs.

B. Half-life and tail

We call half-life the median of T, that is, the time at which half of the systems are still bound. For the tail, we assume survival of the form $S(T) \propto T^{-\alpha}$ above a time $T_{\min}$ and estimate α by maximum likelihood with censoring,

$$\hat{\alpha} = \frac{d}{\sum_{T_i > T_{\min}} \ln(\min(T_i, T_{\max})/T_{\min})}, \tag{1}$$

where d is the number of ejections with $T_i > T_{\min}$ and $T_{\max} = 10^4\, t_{\mathrm{cr}}$; the statistical error is approximately $\hat{\alpha}/\sqrt{d}$. We use $T_{\min} = 100$ and $300\, t_{\mathrm{cr}}$.

C. Excursions

At each recorded instant, we consider the most tightly bound pair and mark an excursion when the third body is more than five semimajor axes of that pair away. For each run, we measure the fraction of the lifetime spent in excursion, the number of excursions, and the duration of the longest one. A long-lived system ($T > 100\, t_{\mathrm{cr}}$) is classified as dominated by a long flight when a single excursion takes up more than half of its lifetime.

III. RESULTS

A. Half-life

Table I summarizes the results. The half-life is 27 to 30 t_{cr} with equal masses, 8 to 9 with two light masses and one heavy mass (1, 1, 10), and about 3 with two heavy masses and one light mass (10, 10, 1). The greater the mass disparity, the shorter the life of the system, in agreement with the result of

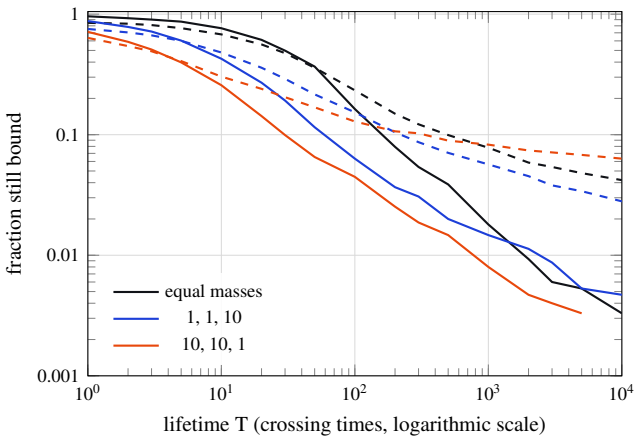


Fig. 1. Fraction of systems still bound as a function of time, on a logarithmic scale on both axes, for the six groups. Solid lines: start from rest; dashed lines: virial start. A power law appears as a straight line.

note FR-TC-05 that a heavy pair ejects the light body after a few encounters [5].

B. Heavy tail

In all groups, long-term survival is many orders of magnitude greater than that predicted by an exponential decay with the same half-life. With equal masses and a start from rest, for example, an exponential with half-life 29.7 would give $S(1000) \approx 10^{-10}$, against an observed 0.018. Fig. 1 shows the survival curves on a logarithmic scale: after a rapid initial drop, they become approximately straight, which indicates a power law. In the start from rest, the exponent lies between 0.63 and 0.94, in the same range as the value 2/3 predicted for Lévy flights [1]. In the virial start, the tail is much heavier, with an exponent between 0.17 and 0.45; with $T_{\min} = 300\,t_{\rm cr}$, the exponents decrease slightly in both sets, which suggests that the tail has not yet reached its final regime within the integrated interval.

C. Longevity mechanisms

The classification of excursions separates two mechanisms. In the start from rest, the long-lived systems are dominated by a single long flight: in 54% (equal masses), 86% (1, 1, 10), and 91% (10, 10, 1) of them, a single excursion takes up more than half of the lifetime, and the median number of excursions is 3 to 11. In the virial start with (10, 10, 1), the picture is different: the long-lived systems make hundreds of short excursions, with a median of 406, and the longest one takes up, in the median, only 3.5% of the lifetime. This is a nearly stable hierarchical configuration, in which the light body repeatedly orbits outside the heavy pair; 6.3% of these systems were still bound at $10^4\,t_{\rm cr}$. The other virial groups lie between the two extremes, with about half of the long-lived systems dominated by a long flight.

IV. DISCUSSION

The half-life describes most systems well, but not the entire statistics: a small and persistent fraction lives much longer than a memoryless process would allow. The results reproduce the algebraic tail described in the literature [1, 2] and show that it has at least two origins. The first is the long flight: an encounter launches a body into a nearly escaping orbit, and the return time, which grows as the binding energy of that orbit becomes very small, can be enormous. The distribution of these return times has a power-law tail, which explains exponents close to 2/3. The second is capture in nearly stable hierarchical configurations, favored when there is angular momentum and a heavy pair: the light body remains in an outer orbit that only breaks up through slow chaotic diffusion, and the tail becomes even heavier.

The difference between the sets is related to angular momentum. In the start from rest, the total angular momentum is zero, and the outer orbits tend to be radial, leading to close encounters at each return; there are no protected hierarchical configurations. In the virial start, angular momentum allows more circular outer orbits, which avoid the inner pair and can last a long time.

The main limitations are the following. The simulations are planar. The limit of $10^4\,t_{\rm cr}$ censors the tail, and the estimated exponent depends on the interval used. The detection of excursions depends on the five-semimajor-axis criterion and on the recorded trajectory, which is decimated in long runs. The classification into two mechanisms is approximate: many systems combine long flights and hierarchical phases.

V. CONCLUSIONS

Three-body systems have a well-defined half-life, from 3 to 30 crossing times depending on the masses, but they do not decay exponentially: from 0.8% to 8% survive to $1000\,t_{\rm cr}$, and the survival tail approximately follows a power law. In the start from rest, the tail comes from long flights, with an exponent close to 2/3; in the virial start, especially with a heavy pair, it also comes from nearly stable hierarchical configurations, which produce even heavier tails.

TRANSPARENCY AND USE OF ARTIFICIAL INTELLIGENCE

The author is a lawyer and not a physicist. This work results from independent research, conducted with the assistance of artificial intelligence (a language model) in the literature searches, the programming of the simulations, the analysis of the results, the writing, and the translation. The author formulated the questions, followed the experiments, and is responsible for the published content.

Table I. Half-life, survival, and tail exponent by mass group and by set (R: start from rest; V: virial), in 1,500 runs per row. $S(T)$ is the fraction of systems still bound at time T, in t_{cr}. α is the tail exponent above $100\,t_{\mathrm{cr}}$, with statistical error.

Masses	Set	Half-life	$S(100)$	$S(1000)$	$S(10^4)$	α
1, 1, 1	R	29.7	0.163	0.018	0.003	0.94 ± 0.06
1, 1, 1	V	26.6	0.233	0.078	0.042	0.45 ± 0.03
1, 1, 10	R	7.7	0.063	0.015	0.005	0.63 ± 0.07
1, 1, 10	V	9.3	0.153	0.057	0.028	0.42 ± 0.03
10, 10, 1	R	3.1	0.045	0.008	0.000	0.79 ± 0.10
10, 10, 1	V	2.8	0.129	0.083	0.063	0.17 ± 0.02

DATA AND CODE AVAILABILITY

All numbers were obtained with scripts published at https://fisica-revisitada.vercel.app/en/code/, which run on Node.js without external dependencies and allow the results to be reproduced.

CONFLICT OF INTEREST

The author declares no conflicts of interest. This work received no funding. This text has not undergone peer review and should be read as a working note. Corrections are recorded in the note itself, with the date.

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